

Robots Kinematics and Robots Pose

Kinematics of robots

- ❑ Kinematics is the study of motion.
- ❑ In robotics it refers to the relationship between **joint movements** and **end effector movements**.
- ❑ Kinematics is the study of the relationship between a robot's **joint coordinates** and its **spatial layout**
- ❑ Kinematics can yield very accurate calculations in many problems, such as positioning a gripper at a place in space, designing a mechanism that can move a tool from point A to point B, or predicting whether a robot's motion would collide with obstacles.



Configuration space and Workspace

A robot's kinematic structure is described by a set of *links*, which for most purposes are considered to be rigid bodies, and *joints* connecting them and constraining their relative movement, for example, **rotational** or **translational joints**.

A robot's layout, at some instant in time, can be described by one of two methods:

- 1. A list of coordinates for each joint** (typically an angle or translation distance) expressed relative to some *reference frame*, aka *zero position*.
- 2. A spatial representation of its links in the 2D or 3D world** in which it operates, e.g., matrices describing the frame of each link relative to some world coordinate system.



Configuration space and Workspace

The list of joint coordinates are known as the ***configuration*** of the robot.

The 2D or 3D world in which the robot lives is known as its ***workspace***.

The importance of a configuration is that it is a ***non-redundant, minimal representation*** of the robot's layout.



Defining robot structures

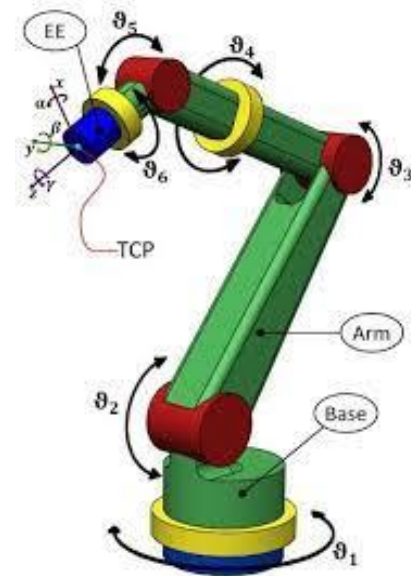
❖ A wide variety of robot mechanisms can be described by categorizing their arrangement of **joints** and **joint types**.

1. Mechanism description by **joint types** each describing the form of relative transformations allowed between the **two links** to which it is attached
2. Mechanisms description by their **topology**, which describes how links and joints interconnect
3. Mechanism description based on whether the robot is affixed to the world or left free to move in space



Categorization based on Joint Types

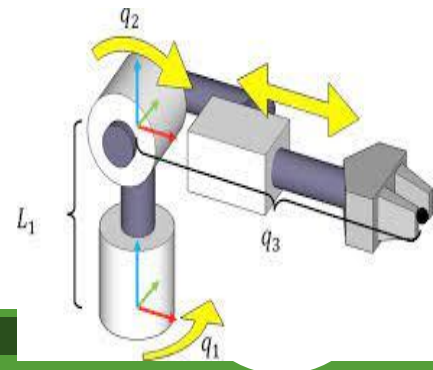
- ❖ There are **three typical joint types**, each describing the form of relative transformations allowed between the **two links** to which it is attached:
- **Revolute (R)**: the attached links rotate about a common axis. A revolute joint (also called pin joint or hinge joint) is a **one-degree-of-freedom** kinematic pair used frequently in mechanisms and machines. The joint constrains the motion of two bodies to pure rotation along a common axis.



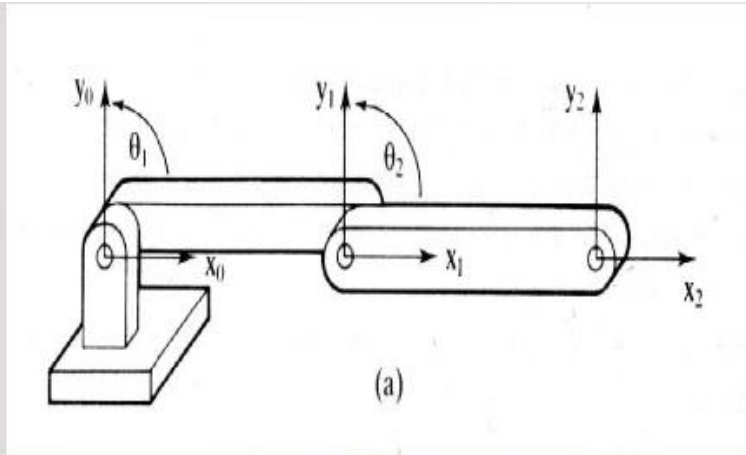
Categorization based on Joint Types

Prismatic (P): The attached links translate about a common axis. A prismatic joint is a connection between two objects that allows relative motion along a single axis. Each joint has **one degree of freedom (1-DOF)**. A prismatic joint allows only linear motion along a single axis.

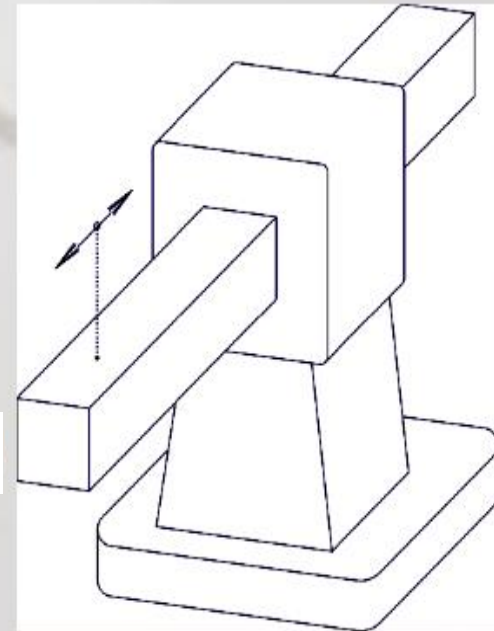
Spherical (S): The attached links rotate about a point. Spherical joints rotate relatively around x, y and z axes, but there will be no relative translation along any of these axes. Each joint has **3-DOF**



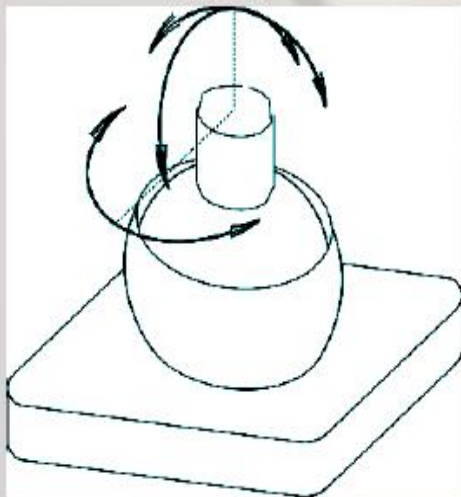
Categorization based on Joint Types (cont)



Revolute Joint



Prismatic Joint

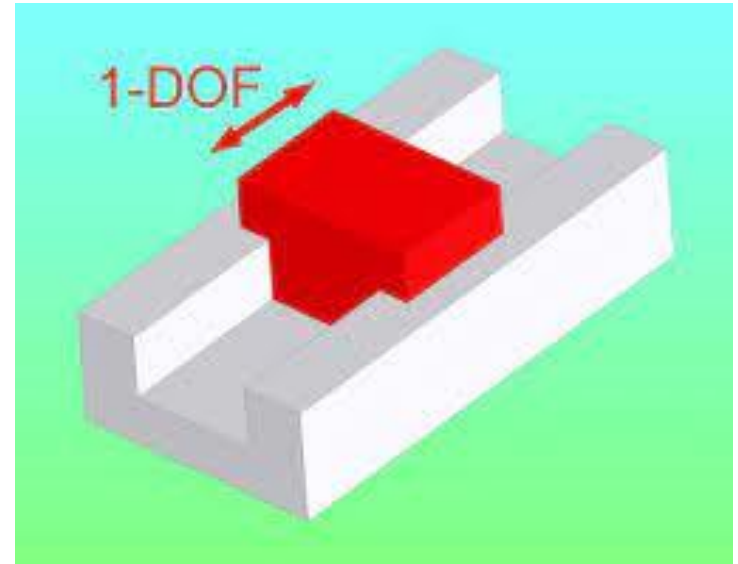


Spherical Joint

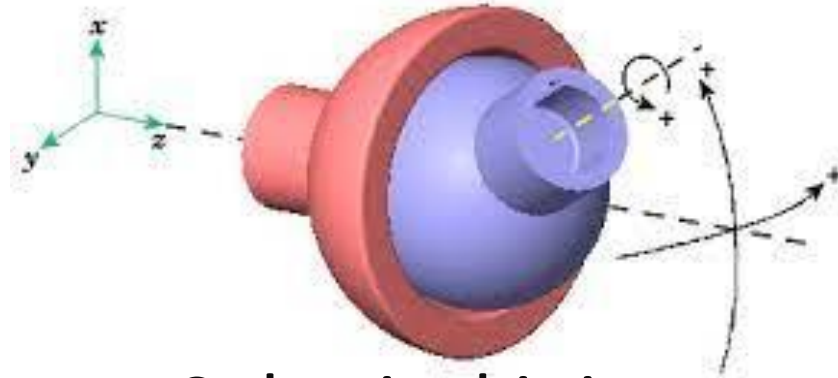
Categorization based on Joint Types (cont)



Revolute joint



Prismatic joint



Spherical joint

Other joints

- Helical (screw) joints
- *Fixed joints* -- attached links are rigidly fixed together
 - since mathematically the two links could be considered as one, this is primarily for representational convenience.
 - We can refer to one of the attached links as the *parent* and the other the *child*.



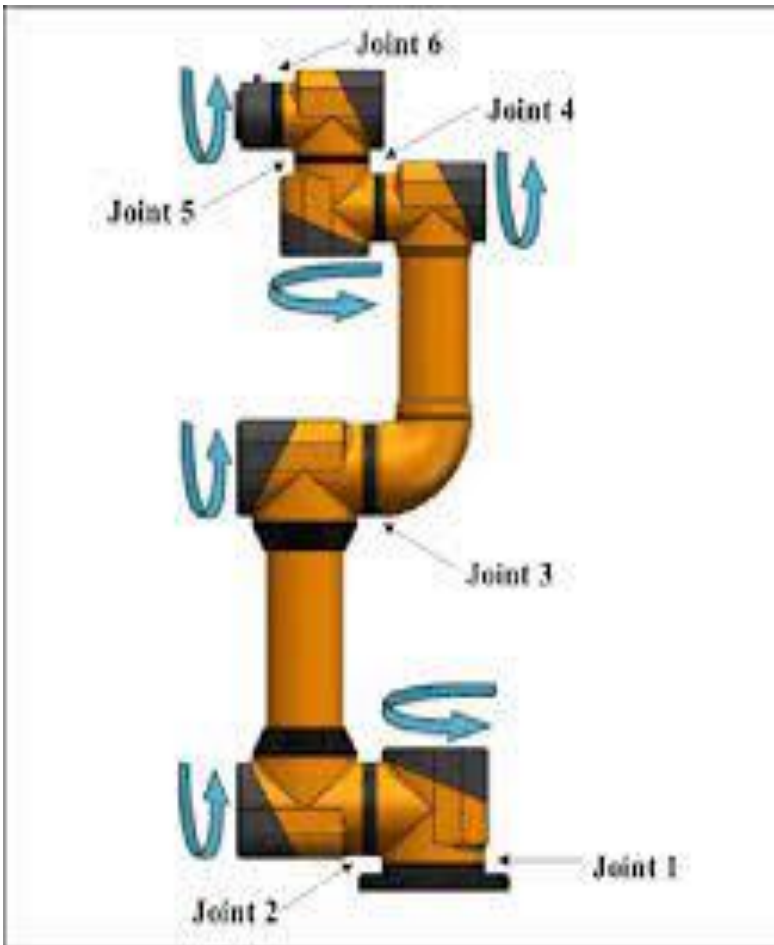
Categorization by joint topology

Mechanisms can be described by their *topology*, which describes how links and joints interconnect:

- **Serial:** the links and joints form a single ordered chain, with the child link of one joint being the parent of the next.
- **Branched:** each link can have zero or more child links, but cutting any joint would detach the system into two disconnected mechanisms. Like a human body, in which fingers are attached to the hand, toes are attached to the feet, and arms, legs, and head are attached to the torso.
- **Parallel:** the series of joints forms at least one closed loop. I.e., there exist joints that, if cut, would not divide the system into two disconnected halves.



Categorization by joint topology (cont)



Serial Joints



Branched Joints



Parallel Joints

Categorization by joint topology (cont)

The topology can be inspected by plotting a *link graph*, which is a network structure in which vertices are links and edges are joints.

Serial mechanisms have a linear link graph, branched mechanisms are trees (i.e., graphs without loops), and parallel mechanisms have loops.

Serial mechanisms are usually characterized using an alphanumeric notation which lists the initials of the joint types in order from the base down the chain. For simplicity, when multiple joints of the same type are repeated, like "XXX", this is listed as "#X" where "#" is the number of repetitions. Examples include:

- 3P3R (PPPRRR): 6-axis CNC machine
- 6R (RRRRRR): revolute joint industrial robot



Categorization based on whether the robot is affixed to the world or left free to move in space

A third characterization defines whether the robot is affixed to the world or left free to move in space:

- **Fixed base:** a base link is rigidly affixed to the world, like in an industrial robot.
- **Floating base:** all links are free to rotate and translate in workspace, like in a humanoid robot.
- **Mobile base:** the workspace is 3D, but a base link can rotate and translate on a 2D plane, like in a car.



Degrees of freedom (DOF) of a robot

- ❖ The degrees of freedom of a robot typically refer **to the number of movable joints of a robot.**
- ❖ A robot with three movable joints will have three axis and three degrees of freedom, a four axis robot will have four movable joints and four axis, and so on
- ❖ Note: A spherical joint has 3-DoF



Degrees of freedom (DOF) of a robot (cont)

The degrees of freedom for a single joint are expressed as the offset of the two attached links from their layout in a given *reference frame*.

For revolute joints, the **one dof (1-DOF)** is a joint angle defining the offset from a joint's zero position along its axis of rotation.

For prismatic joints, the **one dof (1-DOF)** is a translation along the axis relative to its zero position.

For Spherical joints, the **three dof (3-DOF)** can be represented by Euler angles (3 angles represented as (ϕ, θ, ψ)).



Direct or Forward kinematics

- ❖ Forward kinematics is the process of calculating the **frames of a robot's links**, given a configuration and the robot's kinematic structure as input.
- ❖ It refers to the use of the kinematic equations of a robot to compute the **position** of the end-effector from specified values for the joint parameters
- ❖ Eg. To determine **where the robot's hand is?**
(If all joint variables are known)



Direct or Forward kinematics (cont)

The forward kinematics of a robot can be mathematically derived in closed form, which is useful for further analysis during mechanism design, or it can be computed in a software library in microseconds for tasks like motion prediction, collision detection, or rendering.



Direct or Forward kinematics (cont)

Link parameters (link lengths) and joint variables (typically angles) are given and one has to find out the position and orientation of the end-effector (EE).

Given are joint relations (rotations, translations) for the robot arm.

Task: What is the **orientation** and **position** of the end effector?

Forward Kinematics (angles to position)

- What you are given: The length of each link

The angle of each joint

- What you can find: The position of any point (i.e. it's (x, y, z) coordinates



Basic equations – Direct Kinematics (R-P)

- The direct kinematics of the planar *R-P* manipulator is easier to formulate.
- The equations

$$x = d_2 \cos \theta_1 \quad (1)$$

$$y = d_2 \sin \theta_1 \quad (2)$$

$$\varphi = \theta_1 \quad (3)$$

Where: d_2 == the distance between the links

- Equation 1 and 2 represent the **position** of the end effector
- Equation 3 represents the **orientation** of the end effector
- The pose refers to the **position and orientation** of the robot (ie. Equation 1,2 and 3)



Basic equations – Direct Kinematics (3R)

The 3-*R* manipulator

$$x = l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

$$\phi = (\theta_1 + \theta_2 + \theta_3)$$

x , y and φ are the Cartesian coordinates of the end effector representing the **Pose** (**position** and **orientation**) of the robot



Inverse Kinematics

Inverse kinematics is the **mathematical process of calculating the variable joint parameters needed to place the end of a kinematic chain**, such as a robot manipulator or animation character's skeleton, in a given position and orientation relative to the start of the chain.

The reverse process **that computes the joint parameters** that achieve a specified position of the end-effector is known as inverse kinematics.



Inverse Kinematics

Given **link parameters** and **position** and **orientation** of the end effector, one has to find **joint variables**

Inverse kinematics – Given the desired end effector **position** and **orientation**.

Task: What are the **joint rotations and orientations** to achieve this?

Inverse Kinematics (position to angles)

- What you are given: (a) The length of each link (b) The position of some point on the robot
- What you can find: The angles of each joint needed to obtain that position



Inverse Kinematics of R-P manipulator

- The inverse kinematics of the planar *R-P* manipulator is easier to formulate.
- The equations if we restrict the revolute joint to have a joint angle in the interval $[0, 2\pi)$, there are two solutions for the inverse kinematics

$$d_2 = \sigma \sqrt{x^2 + y^2}$$

$$\theta_1 = \text{atan2} \left(\frac{y}{d_2}, \frac{x}{d_2} \right), \text{ where } \sigma = \pm 1$$

Example in class

Note: *Angle in Radians $\times 180^\circ/\pi = \text{Angle in Degrees}$*



Inverse Kinematics for 3R manipulator

- The inverse kinematics analysis for a planar 3-*R* manipulator appears to be complicated but we can derive analytical solutions.
- Recall that the direct kinematics equations for 3-*R* manipulator are:

$$x = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3) \quad (1a)$$

$$y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3) \quad (1b)$$

$$\phi = (\theta_1 + \theta_2 + \theta_3) \quad (1c)$$

- For inverse kinematics we assume that we are given the Cartesian coordinates, x , y , and ϕ and we want to find analytical expressions for the joint angles θ_1 , θ_2 and θ_3 in terms of the Cartesian coordinates.



Inverse Kinematics for 3-R manipulator(cont)

Substituting (1c) into (1a) and (1b) we can eliminate θ_3 so that we have two equations in θ_1 and θ_2

$$x - l_3 \cos \varphi = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) \quad (1d)$$

$$y - l_3 \sin \varphi = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) \quad (1e)$$

*where the unknowns have been grouped on the **right hand side**; the **left hand side** depends only on the end effector or Cartesian coordinates and are therefore known.*



Inverse Kinematics for 3-R manipulator(cont)

Rename the left hand sides, $x' = x - l_3 \cos \varphi$, $y' = y - l_3 \sin \varphi$, for convenience. We regroup terms in (d) and (e), square both sides in each equation and add them:

$$\begin{aligned}(x' - l_1 \cos \theta_1)^2 &= (l_2 \cos(\theta_1 + \theta_2))^2 \\ + \\ (y' - l_1 \sin \theta_1)^2 &= (l_2 \sin(\theta_1 + \theta_2))^2\end{aligned}$$

After rearranging the terms we get a single nonlinear equation in θ_1 :

$$(-2l_1x')\cos\theta_1 + (-2l_1y')\sin\theta_1 + (x'^2 + y'^2 + l_1^2 - l_2^2) = 0 \quad (f)$$



Inverse Kinematics for 3-R manipulator(cont)

Notice that we started with three nonlinear equations in three unknowns in (a-c). We reduced the problem to solving two nonlinear equations in two unknowns (d-e). And now we have simplified it further to solving a single nonlinear equation in one unknown (f).

Equation (f) is of the type

$$P \cos\alpha + Q \sin\alpha + R = 0 \quad (g) \text{ (refer to appendix)}$$

There are two solutions for θ_1 given by:



Inverse Kinematics for 3-R manipulator(cont)

$$\theta_1 = \gamma + \sigma \cos^{-1} \left(\frac{-(x'^2 + y'^2 + l_1^2 - l_2^2)}{2l_1 \sqrt{x'^2 + y'^2}} \right)$$

where,

(h)

$$\gamma = \tan^{-1} 2 \left(\frac{-y'}{\sqrt{x'^2 + y'^2}}, \frac{x'}{\sqrt{x'^2 + y'^2}} \right),$$

and $\sigma = \pm 1$.



Inverse Kinematics for 3-R manipulator(cont)

Note that there are two solutions for θ_1 , one corresponding to $\sigma = +1$, the other corresponding to

$\sigma = -1$. Substituting any one of these solutions back into Equations (d) and (e) gives us:

$$\cos(\theta_1 + \theta_2) = \frac{x' - l_1 \cos \theta_1}{l_2}$$

$$\sin(\theta_1 + \theta_2) = \frac{y' - l_1 \sin \theta_1}{l_2}$$

We can then solve for θ_2 using the atan2 function



Inverse Kinematics for 3-R manipulator(cont)

$$\theta_2 = \text{atan2} \left(\frac{y' - l_1 \sin \theta_1}{l_2}, \frac{x' - l_1 \cos \theta_1}{l_2} \right) - \theta_1 \quad (\text{i})$$

θ_3 can be determined from equation (c)

$$\theta_3 = \phi - \theta_1 - \theta_2 \quad (\text{j})$$

Equations (h-j) are the inverse kinematics solution for the 3-R manipulator



The Pose of a Robot

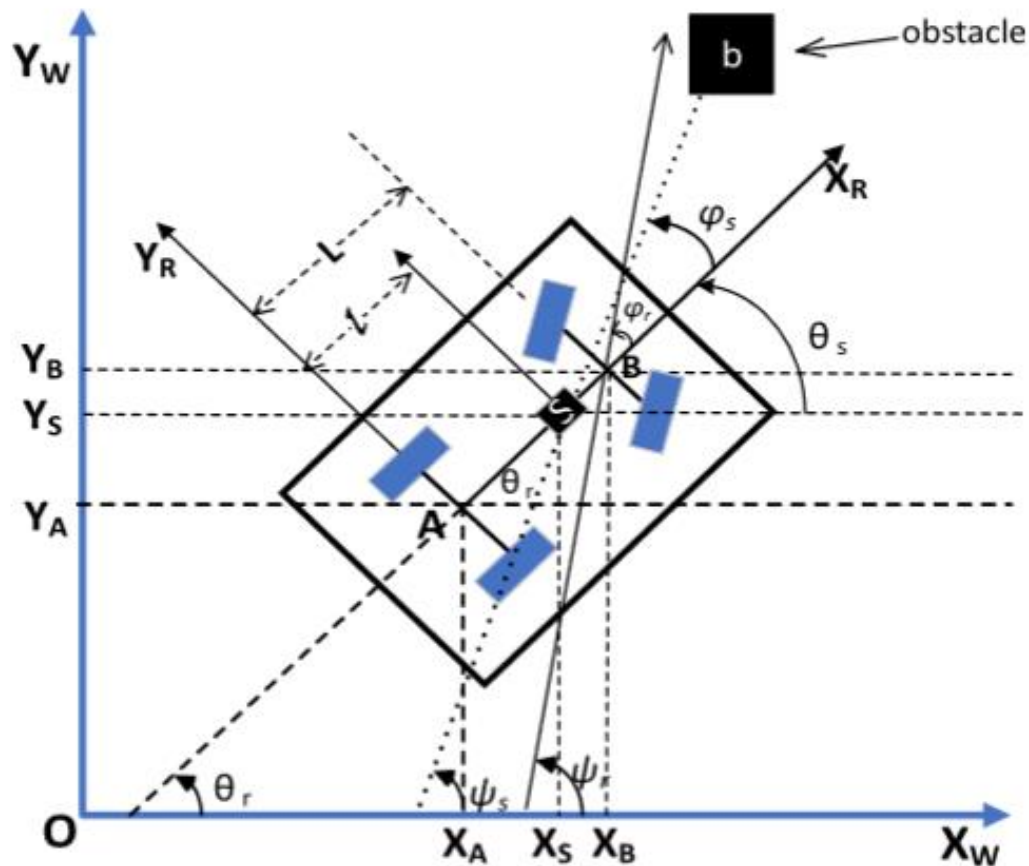


Computing the Pose of a Mobile robot

A car-like vehicle is an example of non holonomic vehicle: all poses can be achieved in the configuration space, but the paths to reach them can be complex



Kinematics model of a car-like mobile robot



Kinematics model of a car-like mobile robot.

Source:

B. B. K. Ayawli et al., Mobile Robot Path Planning in Dynamic Environment Using VD and CG, IEEE Access, 2019



Computing the Pose of a Mobile robot (cont)

The mapping, $\dot{\xi}_R$, between the base and the robot frame can be represented as in (1).

$$\dot{\xi}_R = R(\theta) \cdot [\dot{X}_A, \dot{Y}_A, \dot{\theta}_r]^T \quad (1)$$

where $R(\theta)$ is given as in (2)

$$R(\theta) = \begin{bmatrix} \cos(\theta_r) & \sin(\theta_r) & 0 \\ -\sin(\theta_r) & \cos(\theta_r) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

Two nonholonomic constraints can be obtained as given in (3) and (4) .

$$-\dot{X}_A \sin \theta_r + \dot{Y}_A \cos \theta_r = 0 \quad (3)$$

$$-\dot{X}_B \sin(\theta_r + \varphi_r) + \dot{Y}_B \cos(\theta_r + \varphi_r) = 0 \quad (4)$$

X_B and Y_B represent the location coordinates of the front wheel midpoints B . Hence X_B and Y_B can be computed as in (5) and (6).

$$X_B = X_A + L \cos \theta_r \quad (5)$$

$$Y_B = Y_A + L \sin \theta_r \quad (6)$$



Computing the Pose of a Mobile robot (cont)

where L denotes the distance from midpoint A of the rear wheel axis to midpoint B of the front wheel axis. Based on (5) and (6), the nonholonomic constraints given in (4) can be written as in (7).

$$-\dot{X}_B \sin(\theta_r + \varphi_r) + \dot{Y}_B \cos(\theta_r + \varphi_r) + L(\cos \varphi_r)\dot{\theta}_r \quad (7)$$

Based on (5) and (6), we can obtain the pose of the sensor $S = [x_s, y_s, \theta_s]^T$ using (8).

$$S = \begin{bmatrix} x_s \\ y_s \\ \theta_s \end{bmatrix} = \begin{bmatrix} X_A + l \cos \theta_r \\ X_A + l \sin \theta_r \\ \theta_r \end{bmatrix} \quad (8)$$

where l represents the perpendicular distance between the midpoint of the sensor S and the midpoint A of the rear wheels.



Computing the Pose of a Mobile robot (cont)

Hence the pose of a mobile robot can be computed using eq. 8

$$\bar{P} = \begin{bmatrix} \bar{x} \\ \bar{y} \\ \bar{\theta} \end{bmatrix} = \begin{bmatrix} x + L\cos\theta \\ y + L\sin\theta \\ \theta \end{bmatrix}, \text{ where } L=\text{distance}$$

Example

1. A car-like mobile robot navigated from point A to B covering a distance of 20 meters. If the pose of the robot at point A is $P = (2, 5, 30^\circ)^T$, compute the current pose $\bar{P} = (\bar{x}, \bar{y}, \bar{\theta})^T$ of the robot at point B.

$$P = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 30^\circ \end{bmatrix}, \quad L=20 \text{ meters}$$

$$\bar{P} = \begin{bmatrix} \bar{x} \\ \bar{y} \\ \bar{\theta} \end{bmatrix} = \begin{bmatrix} x + L\cos\theta \\ y + L\sin\theta \\ \theta \end{bmatrix}$$

$$\bar{P} = \begin{bmatrix} 2 + (20 * \cos 30^\circ) \\ 5 + (20 * \sin 30^\circ) \\ 30^\circ \end{bmatrix} = \begin{bmatrix} 2 + (20 * 0.8660) \\ 5 + (20 * 0.5) \\ 30^\circ \end{bmatrix} = \begin{bmatrix} 19.32 \\ 15 \\ 30^\circ \end{bmatrix}$$

$$\bar{P} = (19.32, 15, 30^\circ)^T$$





Appendix 1

Holonomic constraints— A geometric constraint expressed through **positional** variables. It limit the motion of the system to a manifold of the configuration space depending on the initial condition

Non holonomic constraints -- A geometric constraint expressed through **derivatives of positional variables**. It does not limit the accessible configurations, but limits the paths that can be followed to reach them



Appendix 2

A car-like vehicle is an example of non holonomic vehicle: all poses can be achieved in the configuration space, but the paths to reach them can be complex

